

Big Data Systems Project Proposal

Personalized Recipe Recommendation System

Team: Bias & Variance

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1. Project Abstract

As digital recipe platforms grow, users often face difficulty deciding what to cook due to the overwhelming number of available recipes. Traditional search or tag-based filtering methods do not effectively capture personal taste preferences. This project proposes a personalized recipe recommendation system that analyzes large-scale recipe and user interaction data to recommend recipes tailored to individual preferences.

The system will utilize the Food.com dataset containing multiple recipes and user interactions. Using big data processing techniques and machine learning models, the project will identify patterns in user ratings and preferences to generate personalized recommendations. The system will learn hidden relationships between users and recipes and suggest recipes that best match a user's taste profile.

By leveraging big data technologies and machine learning algorithms, the proposed system aims to improve user experience by reducing decision fatigue and helping users discover recipes aligned with their preferences.

2. Problem Statement and Objectives

Problem Statement

Online recipe platforms contain large collections of recipes with diverse ingredients, cooking styles, and preparation methods. As users add more recipes to their personal libraries, it becomes increasingly difficult to select recipes that match their preferences and dietary habits. Existing platforms primarily rely on manual browsing, search filters, or basic sorting mechanisms, which do not adapt to individual taste patterns.

The challenge is to develop a system capable of analyzing large volumes of recipe and user interaction data to automatically learn user preferences and generate accurate personalized recommendations. This requires processing large datasets, identifying behavioral patterns, and building machine learning models that can predict which recipes a user is likely to prefer.

Objectives

1. Data Collection and Preparation

Collect and clean recipe and interaction data from the Food.com dataset to create a structured dataset suitable for analysis and model training.

2. Feature Engineering

Extract important features from recipes such as cooking time, ingredients, nutritional information, and tags to enhance model performance.

3. Model Training

Develop and train machine learning models capable of learning user taste patterns from historical recipe ratings and interactions.

4. **Recommendation Generation**

Implement algorithms that recommend recipes to users based on predicted preferences.

Scalability

Ensure that the system can efficiently process large datasets containing hundreds of thousands of recipes and millions of interactions.

5. **Accuracy and Personalization**

Improve recommendation accuracy by learning latent preference patterns among users and recipes.

6. **Evaluation**

Evaluate the model using metrics such as prediction accuracy and recommendation relevance.

7. **System Improvement**

Continuously refine the recommendation system by incorporating additional interaction data and retraining the model when necessary.

3. **Data Source**

Dataset Name: Food.com Recipes and User Interactions

Source: Kaggle, kaggle.com/datasets/shuyangli94/food-com-recipes-and-user-interactions

Lineage: Scraped from Food.com; published by Majumder, Li, Ni, McAuley (EMNLP 2019). Real user ratings collected over 18 years (2000-2018). Licensed under CC BY-SA 3.0 with 400+ academic citations.

Files Used:

- RAW_recipes.csv — 231,637 rows; columns used: id, name, tags, minutes, nutrition
- RAW_interactions.csv — 1,048,576 rows; columns used: user_id, recipe_id, rating (1-5 stars)

Approximate Number of Records: ~1.28 million total records (231,637 recipes + 1,048,576 interactions)

Alignment with Big Data Project: The 1M+ interaction records justify distributed processing with PySpark and Spark MLlib's ALS implementation. Recipe tags bridge the Food.com training data to Mealie's recipe library, enabling generalized recommendations without requiring ID-level matching. New Mealie events will be streamed via Kafka and appended to the training store.

System Architecture Overview

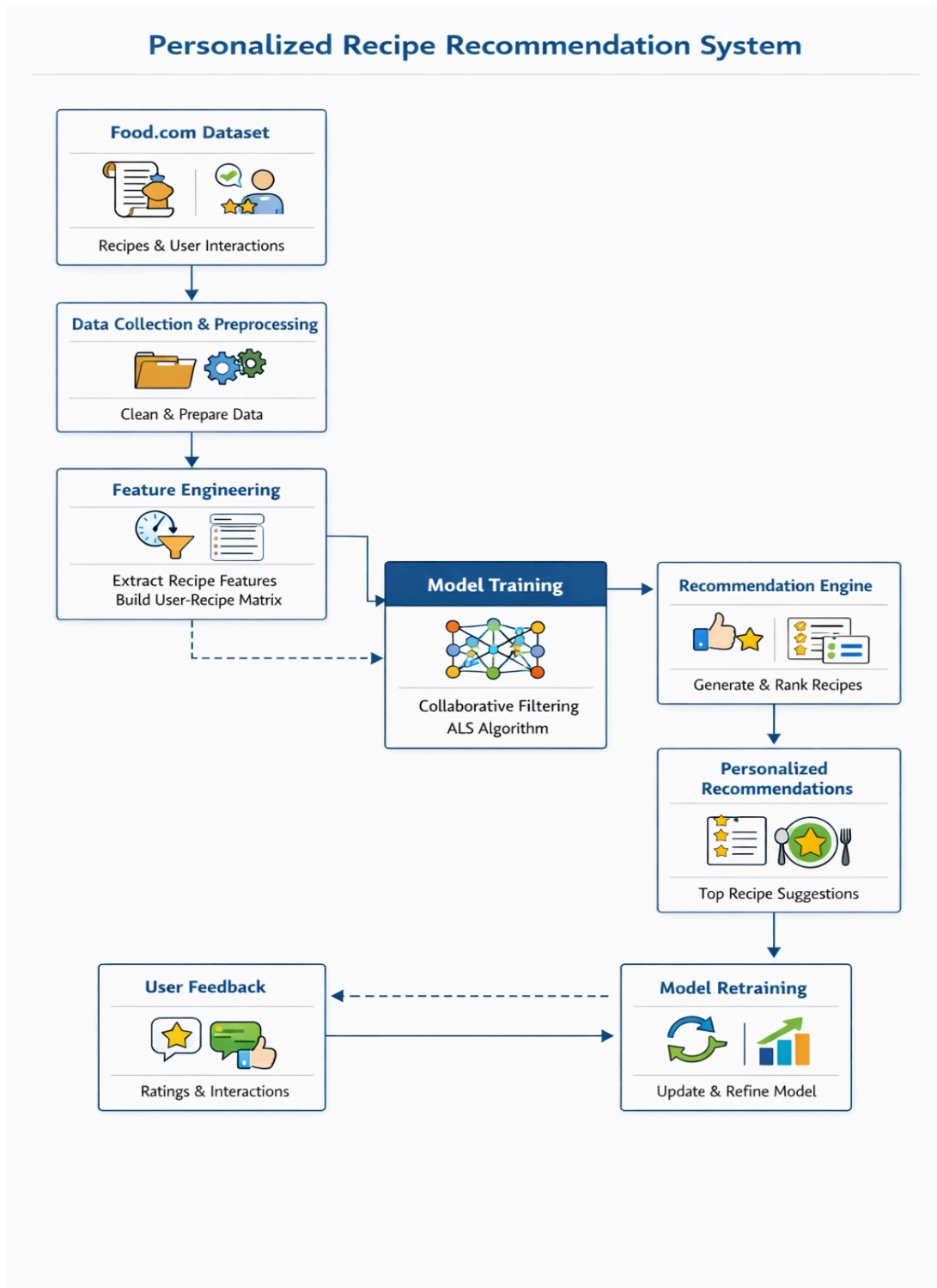


Figure 1: Personalized Recipe Recommendation System — End-to-End Data Flow

4. Proposed Technologies and Programming Languages

Python is the programming language used throughout. The table below maps each tool to its specific role in the pipeline.

Pipeline Stage	Technology	Usage
Data Ingestion	Apache Kafka	Streams user rating events and new recipes added in Mealie into the pipeline in real time.
Storage	HDFS	Stores raw CSVs, processed Parquet files for faster reads, and the 50-dim recipe vectors are also persisted here partitioned by recipe_id for faster lookups.
Preprocessing	PySpark	All cleaning and transformation runs on Spark. This includes removing users with fewer than 5 ratings, normalizing rating values, and joining the recipe metadata with the interactions table across 1.28M records
Model Training	Spark MLlib (ALS)	Trains a distributed matrix factorization model on the 1M+ rating matrix to generate user and recipe vectors. Each user and recipe is represented as a 50-dimensional latent vecto
Evaluation	Spark MLlib Metrics	Computes RMSE and Precision@K on a held-out test split.
Visualization	Matplotlib / Seaborn	Plots rating distributions and top-N recommendation results.

Conclusion

This project proposes a big data–driven personalized recipe recommendation system using the Food.com Recipes and User Interactions dataset. By analyzing large-scale user ratings and recipe metadata, the system aims to learn user taste preferences and generate relevant recipe recommendations. Machine learning techniques such as collaborative filtering will help uncover hidden relationships between users and recipes, enabling the system to suggest recipes that align with individual preferences.

The proposed system demonstrates how big data technologies and machine learning can enhance user experience on recipe platforms by reducing decision fatigue and improving recipe discovery. With scalable data processing tools and continuous model retraining using user feedback, the recommendation system can adapt to changing user preferences over time and provide increasingly accurate recommendations.